Morgan Stanley Services Canada Corporation

November 30, 2025

Morgan Stanley
700 Wellington St.
Montreal, QC, Canada H3C 3S4

Internship Application for Summer 2026 Quantitative Finance Program in Montreal

To whom it may concern,

About Me

Thank you for considering my application. I am currently a PhD candidate in experimental quantum optics at the University of Toronto under Prof. Joseph Thywissen. My interest in financial markets developed early in my PhD during the post-COVID-19 period of heightened market volataility and intense focus on interest-rate policy. Witnessing the economic dynamism of that period inspired me to pursue quantitative finance as a career path, especially in Fixed Income or macroeconomics, where analytical insights can drive decisions. As I approach completion of my program, I am eager to apply my rigorous training in research, data analysis, and statistical modeling to an exciting and innovative field.

Why Morgan Stanley?

I am drawn to Morgan Stanley for its leadership in the industry and its reputation for excellence. I first learned about the Summer Internship Program through a senior graduate student in my research group, whose highly positive experience ultimately led to a full-time role at the firm. His endorsement played a significant role in my decision to apply, and it highlighted to me the strength of Morgan Stanley's culture and the quality of its training.

Why Me?

I believe my experience as an experimental physicist has prepared me well for quantitative finance. My research requires a multidisciplinary skill set that emphasizes analytical thinking, adaptability, and rapid problem-solving. I look forward to the opportunity to contribute my skills at Morgan Stanley this summer and to learn from the firm's talented professionals.

Sincerely,

Kevin Xie

Attached: Résumé

Kevin Xie

PhD Candidate · Physics · University of Toronto

(+1) 416-993-6322 | kevin.gs.xie@gmail.com | Kevin G. S. Xie

Summary

PhD candidate in experimental quantum optics with 8+ years of experience spanning physics research, engineering, and industry R&D. Strong foundation in quantitative analysis, modeling, and development. Motivated by learning, adapting, and applying technical skills to new domains, including finance.

Experience

University of Toronto (Prof. Joseph Thywissen) — *Toronto, Canada*

PhD Researcher, Experimental Quantum Optics — *May 2020 – Present*

- **Research focus**: precision measurements, high-dimensional data analysis, control of complex physical systems.
- Conducted experiments on ultracold quantum gases, including spectroscopy of strongly interacting fermions to study many-body phenomena.
- Applied model fitting, parameter estimation, and uncertainty analysis to extract physical parameters from noisy data.
- Deployed and maintained custom control, real-time monitoring, and automated data-acquisition systems (Python, C++).
- Led a research team of four, directing experimental strategy, coordinating measurement protocols, and managing data-analysis pipelines.
- Regularly communicated results to international collaborators; presented findings at conferences and peer-reviewed journals.

University of Waterloo (Prof. Jan Kycia) — *Waterloo, Canada*

Undergraduate Researcher — *Jan. – Sep. 2017, Sep. 2019 – Apr. 2020*

- Engineered and analyzed thermal properties of components for a hybrid Helium-3 dilution refrigerator aimed at millikelvin-scale spin-qubit characterization.

Lumentum — *Ottawa, Canada*

Optical Algorithms Developer — *Sep. – Dec. 2018, May – Aug. 2019*

- Built a predictive failure-detection neural network model (>96% accuracy) to optimize calibration of LCoS components in optical networking devices, aimed at improving throughput and automation.
- Standardized test-station infrastructure across multiple facilities, eliminating bottlenecks and increasing throughput by up to 10%.
- Conducted comprehensive performance studies on next-generation bandwidth-enhancement algorithms for optical module R&D.
- Replaced outdated internal component data retrieval tool with modern Python-based solution adopted company-wide.

TRIUMF (T2K Canada) — *Vancouver, Canada*

Undergraduate Researcher — *Sep. 2017 - Apr. 2018*

- Detected and corrected periodic temperature fluctuations that introduced systematic error in photomultiplier-tube characterization. Validated and reproduced by international collaborators (KEK).

Extracurricular Activities

Kaggle, Jane Street Group — *Online*

Competition Entrant — *2024*

- Applied quantitative finance principles and machine learning algorithms (PyTorch, TensorFlow) to large-scale financial time-series data; developed feature engineering and model optimization with high-performance data tools (Polars).

University of Toronto Graduate Students' Union — *Toronto, Canada*

Member of Board of Directors and Elections & Referenda Committee — *Oct. 2023 – Apr. 2024*

- Organized and supported policy development and governance initiatives affecting 20,000+ graduate students, contributing to improvements in academic, financial, and community resources.

The 27th International Conference on Atomic Physics (ICAP 2022) — *Toronto, Canada*

Volunteer — *July 2022*

- Assisted in organizing and running an international conference with 500+ participants, providing on-site logistics, A/V support, and management of virtual conference components.

Skills

Programming	Python (NumPy, SciPy, Pandas), MATLAB, Mathematica, C++, Linux, Git, LaTeX
Data	Uncertainty Analysis, Optimization, Signal Processing, Machine Learning (PyTorch, TensorFlow)
Technical Systems	Experimental Control Systems, Instrument Automation
Languages	English (native), Cantonese (beginner/intermediate)
Interests	Music (Jazz), Boxing, Macroeconomics, Geopolitics, History

Education

University of Toronto *Toronto, Canada*

PhD Physics *Sep. 2020 - Present*

University of Waterloo *Waterloo, Canada*

Bsc Physics, Co-op *Sep 2015 - Apr 2020*

Publications

* indicates shared first authorship.

Published

C. J. Dale*, **K. G. S. Xie***, K. Pond Grehan, S. Zhang, J. Maki, and J. H. Thywissen. Emergent *s*-wave interactions in orbitally active quasi-two-dimensional Fermi gases. Phys. Rev. A 110, L051302 (2024)

K. G. Jackson, C. J. Dale, J. Maki, **K. G. S. Xie**, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent *s*-wave interactions between identical fermions in quasi-one-dimensional geometries. Phys. Rev. X 13, 021013 (2023)

C. R. C. Buhariwalla, Q. Ma, L. DeBeer-Schmitt, **K. G. S. Xie**, D. Pomaranski, J. Gaudet, T. J. Munsie, H. A. Dabkowska, J. B. Kycia, and B. D. Gaulin. Long-wavelength correlations in ferromagnetic titanate pyrochlores as revealed by small-angle neutron scattering. Phys. Rev. B 97, 224401 (2018)

In Review

K. G. S. Xie*, C. J. Dale*, K. Pond Grehan, M. F. Wang, T. Enss, P. S. Julienne, Z. Yu, and J. H. Thywissen. Dimer-projection contact and the clock shift of a unitary Fermi gas. (under review)

Presentations

Contributed Talks and Posters

K. G. S. Xie, C. J. Dale, K. Pond Grehan, M. F. Wang, Z. Yu, P. S. Julienne, T. Enss, and J. H. Thywissen. Clock shift and contact of a unitary Fermi gas inferred by dimer association. Oral presentation: DAMOP 2025, Portland, Oregon, U.S.A.

K. G. S. Xie, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent *s*-wave dimers near a *p*-wave Feshbach resonance in a strongly confined Fermi gas. Oral presentation and poster: ICPEAC 2023, Ottawa, Ontario, Canada.

K. G. S. Xie, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, S. Zhang, and J. H. Thywissen. Associating emergent *s*-wave dimers along strongly confined directions of a spin-polarized Fermi gas. Oral presentation: DAMOP 2023, Spokane, Washington, U.S.A.

K. G. S. Xie, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Feshbach-enhanced odd- and even-wave interactions between identical fermions in quasi-one-dimensional confinement. Poster: ICAP 2022, Toronto, Ontario, Canada.

K. G. S. Xie, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Kokkelmans, S. Smale, P. S. Julienne, and J. H. Thywissen. *P*-wave interactions in a spin-polarized Fermi gas with and without confinement. Poster: DAMOP 2022, Orlando, Florida, U.S.A.

Honors & Awards

2025	**SGS Conference Travel Grant,** University of Toronto	*Toronto, Canada*
2023	**Ontario Graduate Scholarship,** University of Toronto	*Toronto, Canada*
2021	**Canada Graduate Research Scholarship – Master's,** NSERC	*Toronto, Canada*
2020	**Dean's Honours List,** University of Waterloo	*Waterloo, Canada*
2017	**Undergraduate Student Research Award,** NSERC	*Waterloo, Canada*
2016	**President's Scholarship of Distinction,** University of Waterloo	*Waterloo, Canada*

Teaching Experience

2025	**PHY324,** Teaching Assistant	*Toronto, Canada*
2025	**PHY293,** Teaching Assistant	*Toronto, Canada*
2024	**PHY294,** Teaching Assistant	*Toronto, Canada*
2024	**PHY405,** Teaching Assistant	*Toronto, Canada*
2023	**PHY424,** Teaching Assistant	*Toronto, Canada*
2021, 2022	**PHY224,** Teaching Assistant	*Toronto, Canada*
2020, 2021	**PHY131,** Teaching Assistant	*Toronto, Canada*